TASK23: AI PROMPT APP

AIM: To develop an application that demonstrates prompt engineering by generating and evaluating code or cloud configurations based on user input.

ALGORITHM:

1. Start the program
2. Create project files (HTML, CSS, JS, server.js, db.js)
3. Design UI with input field for user prompt
4. Capture user prompt using JavaScript
5. Send prompt to server using fetch API
6. Receive prompt data in backend (server.js)
7. Process prompt using predefined logic (db.js)
8. Generate output (code / cloud config / response)
9. Evaluate output quality and assign score
10. Send generated output and score to frontend
11. Display output and score on UI
12. Stop the program

PROGRAM:

INDEX.HTML:

```
<!DOCTYPE html>
<html>
<head>
   <title>AI Prompt App</title>
   <link rel="stylesheet" href="style.css">
</head>
<body>
```

```
<div class="container">
    <h2>Prompt Engineering Tool</h2>
    <textarea id="prompt" placeholder="Enter your prompt..."></textarea>
    <button onclick="generate()">Generate Output</button>
    <h3>Generated Output</h3>
    <pre id="output"></pre>
    <h3>Quality Score</h3>
    <p id="score"></p>
</div>
<script src="script.js"></script>
</body>
</html>
```

STYLE.CSS:

```
body {
    font-family: Arial;
    background: #202038;
    color: white;
    display:flex;
    justify-content:center;
    align-items:center;
    height:100vh;
}
.container {
    width: 400px;
    text-align: center;
}
```

```
textarea {
    width: 100%;
    height: 100px;
    margin: 5px;
}
button {
    padding: 10px;
    margin: 10px;
    background: #00c6ff;
    border: none;
    color: white;
}
pre {
    background: #333;
    padding: 10px;
}
```

SCRIPT.JS:

```
function generate() {
    const prompt = document.getElementById("prompt").value;
    fetch("/generate", {
        method: "POST",
        headers: {"Content-Type":"application/json"},
        body: JSON.stringify({ prompt })
    })
    .then(res => res.json())
    .then(data => {
```

```
        document.getElementById("output").innerText = data.output;
        document.getElementById("score").innerText = data.score + "/10";
    });
}
```

SERVER.JS:

```
const express = require("express");
const db = require("./db");

const app = express();

// Middleware
app.use(express.json());
app.use(express.static(__dirname)); // serve HTML

// Home route (optional but useful)
app.get("/", (req, res) => {
    res.sendFile(__dirname + "/index.html");
});
// Generate output
app.post("/generate", (req, res) => {
    const result = db.processPrompt(req.body.prompt);
    res.json(result);
});
// START SERVER WITH LINK
app.listen(3000, () => {
    console.log(" 🚀 Server running at:");
```

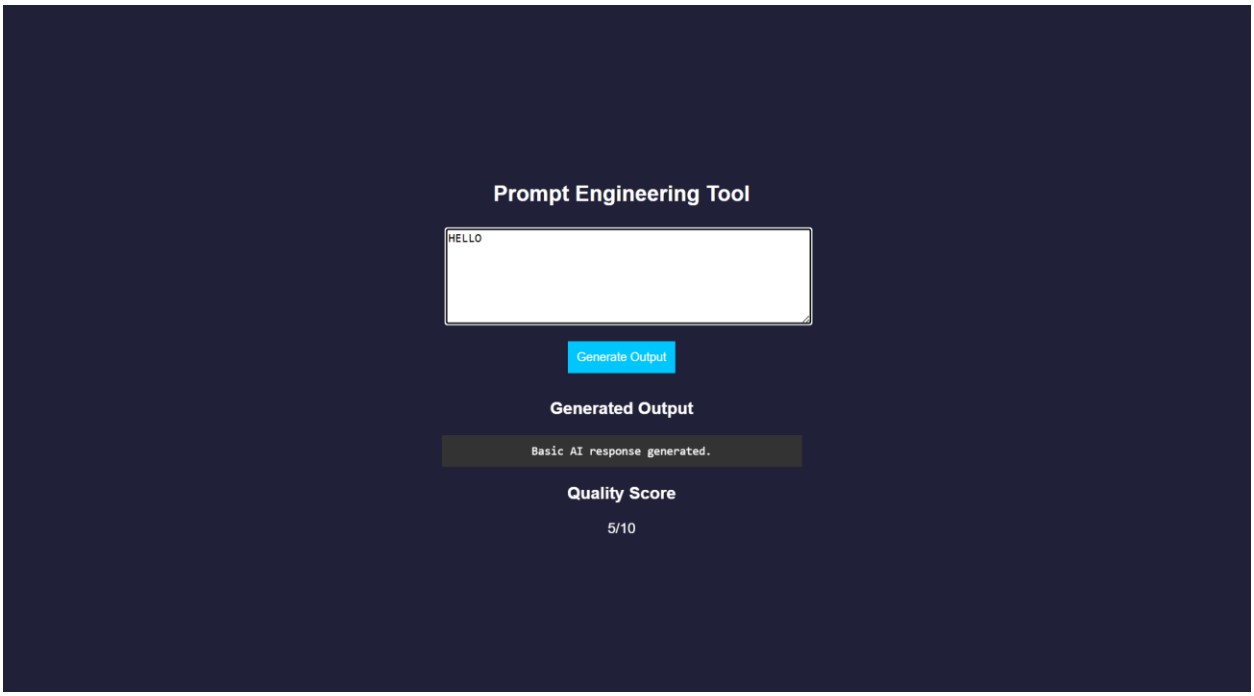
```
    console.log(" http://localhost:3000");
});
```

DB.JS:

```
function processPrompt(prompt) {
    let output = "";
    let score = 5;
    if (prompt.toLowerCase().includes("code")) {
        output = `function greet() {
    console.log("Hello from AI!");
}`;
        score = 9;
    }
    else if (prompt.toLowerCase().includes("cloud")) {
        output = `AWS EC2 Config:
- Instance: t2.micro
- Region: us-east-1`;
        score = 8;
    }
    else {
        output = "Basic AI response generated.";
        score = 5;
    }

    return { output, score };
}
module.exports = { processPrompt };
```

OUTPUT:



RESULT: A prompt engineering application was successfully developed to generate and evaluate AI-based outputs for code and cloud configurations.